NAME

mbpy_plot − Create an executable shellscript which will generate a GMT modern-mode map of swath sonar data.

VERSION

Version 5.0

SYNOPSIS

mbpy_plot [**−I***file*] [**−F***format*] [**−O***root*] [**−−format** *fmt*] [**−−title** *text*] [**−−colorbar−label** *text*] [**−−dpi** *dpi*] [**−X**] [**−V**]
[**−R***west/east/south/north*] [**−J***spec*] [**−P***pagesize*] [**−−portrait** | **−−landscape**] [**−−scale−location** *l|r|t|b*] [**−−tick−info** *spec*] [**−−length−scale** *spec*] [**−−timestamp** [*label*]]
[**−G***color_mode*] [**−−filtered**] [**−W***color_style/palette*[*/ncolors*] | *cptfile*] [**−D**] [**−Y**] [**−S**] [**−Z***min/max*]
[**−A***magnitude*[*/azimuth* | */zero_level*]] [**−−magnitude** *val*] [**−−azimuth** *deg*] [**−−zero−level** *val*] [**−−shade−flip**] [**−−stretch−shade**]
[**−C**[*cont_int/color_int/tick_int/label_int/tick_len/label_hgt*]] [**−−contour−algorithm** *n*] [**−−contour−nplot** *pings*]
[**−N**[*time_tick/time_annot/date_annot/tick_len* | **F** | **FP**]] [**−−nav−names** [*height*[*/P*]]] [**−−ping−ticks** [*tick/annot/len*]]
[**−−footprint** *factor/mode/depth*] [**−−swath−scale** *mode/scale*[*/min/max*]] [**−−begin−time** *yr/mo/da/hr/mn/sc*]
[**−−end−time** *yr/mo/da/hr/mn/sc*] [**−−speedmin** *speed*] [**−−timegap** *minutes*] [**−−lonflip** *−1|0|1*]
[**−T**] [**−−coast−resolution** *res*] [**−−coast−dry−fill** *color*] [**−−coast−wet−fill** *color*] [**−−coast−lake−fill** *color*]
[**−−coast−river** *spec*] [**−−coast−boundary** *spec*] [**−−coast−pen** *pen*]
[**−−xy−overlay** *file*[*:fill=..*][*:symbol=..*][*:pen=..*]] [**−−text** *lon/lat/size/angle/font/justify/text*] [**−−gmt−default** *param/value*]

DESCRIPTION

mbpy_plot is a Python/PyGMT-era replacement for the Perl macro **mbm_plot**. It reads a swath sonar data file, or a datalist of them, and generates an executable shellscript of **GMT modern mode** commands which, when run, maps the swath data themselves (rather than a grid made from them) with the **MB-System** GMT modules **mbswath** and **mbcontour**. The map may include a bathymetry color fill (**−G1**), bathymetry color shaded relief (**−G2**), bathymetry shaded by beam amplitude (**−G3**), an amplitude fill (**−G4**) or a sidescan fill (**−G5**); bathymetry contours (**−C**); the ship track with time and file name annotation (**−N**); a coastline; xy data drawn as lines or symbols; text labels; a color scale; and a distance scale. If none of **−G**, **−C** and **−N** is given, the navigation track is plotted.

The map bounds come from the files' ".inf" files (made by **mbinfo** or **mbdatalist −O**) when they exist and no time window is given, and otherwise from running **mbinfo**; they are padded by 5% and, if needed, widened so that the map is no more than twice as long as it is wide. The map is scaled to fit the page size (**−P**) in the orientation that gives the larger map, unless the projection or scale is given with **−J**. Page layout and projection fitting are shared with **mbpy_grdplot**.

The generated script is named *root*.sh and makes *root*.*fmt* (**pdf** by default). A generated color palette is written to *root*.cpt (and, for a stretched amplitude shade, *root*.shade.cpt) by **mbpy_plot** itself. When the input is a single swath file rather than a datalist, the script makes and then removes a one-line datalist *root*_datalist.mb−1 for it. No gmt.conf file is created or deleted in the current directory.

As in **mbpy_grdplot**, the colon-separated **−M***sub* options of **mbm_plot** are replaced by descriptive long options. Palettes are limited to 1-7, as in **mbm_plot**.

mbpy_plot requires **GMT** 6 or later with the **MB-System** GMT modules, and a Python 3 interpreter. It uses the modules **mbpy_layout.py**, **mbpy_color.py**, **mbpy_misc.py**, **mbpy_grdplot_script.py** and

mbpy_plot_script.py, installed alongside it.

MB-SYSTEM AUTHORSHIP

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OPTIONS

−I, **−−input** *file*
Swath data file or datalist. Default: datalist.mb−1.

−F, **−−data−format** *format*
MB-System format id of the input file, or −1 for a datalist. Default: determined with **mbformat**.

−O, **−−output** *root*
Output root name. Default: the input's basename without its extension.

−−format *fmt*
Output format passed to **gmt begin**: pdf, png, ps, jpg, tif, eps, ... Default **pdf**.

−−title *text*
Map title. Default: "Data List File *file*" or "Data File *file*".

−−colorbar−label *text*
Color scale label. Default: "Depth (meters)", "Beam Amplitude Values" or "Sidescan Pixel Values", by color mode.

−−dpi *dpi*
Resolution of the **mbswath** image (**mbswath −E**) and of a raster output file.

−X, **−−execute**
Run the generated script immediately after writing it.

−V, **−−verbose**
Print progress and a summary of the map.

−R, **−−region** *west/east/south/north*
Map bounds in longitude and latitude. Default: the data bounds, padded as described above.

−J, **−−projection** *spec*
GMT map projection with a scale or width, e.g. **m1.0** or **M6i**; giving only the projection lets **mbpy_plot** choose the scale. Default: a Mercator projection scaled to fill the page.

−P, **−−pagesize** *pagesize*
Page size code (a, b, c, d, e, f, e1, a0-a10, b0-b10, c0-c7, m1-m6), as in **mbm_plot**. Default **a**.

−−portrait, **−−landscape**
Force the page orientation instead of choosing it automatically.

−−scale−location *l|r|t|b*
Place the color scale at the left, right, top or bottom of the map. Default **b**.

−−tick−info *spec*
Raw **−B** basemap tick-interval spec, replacing the computed one.

−−length−scale *spec*
Draw a distance scale with **gmt basemap −L***spec*, e.g. "jBL+c36.8+w5k+o0.3/0.3". The word "center" in *spec* is replaced by the longitude/latitude of the map center, e.g. "gcenter+c36.8+w5k".

−−timestamp [*label*]
Add a GMT time stamp (**−U**), with an optional label.

−G, **−−color−mode** *color_mode*
Swath image style (**mbswath −Z**):
color_mode = 1: Bathymetry color fill.
color_mode = 2: Bathymetry color shaded relief.
color_mode = 3: Bathymetry color shaded by beam amplitude.
color_mode = 4: Beam amplitude fill.
color_mode = 5: Sidescan fill.

−−filtered
Plot filtered amplitude or sidescan values (**mbswath −Z***mode***F**).

−W, **−−color−control** *color_style/palette*[*/ncolors*] | *cptfile*
Color palette. *color_style* 1 gives a continuous palette, 2 a discrete one. *palette* is 1 Haxby colors, 2 high intensity colors, 3 low intensity colors, 4 grayscale, 5 uniform gray, 6 uniform black or 7 uniform white; no color scale is drawn for 5-7. *ncolors* is the number of colors [11]. Default: 1/1, or 1/4 (grayscale) for color modes 4 and 5. If the argument names an existing file, that file is used as the color palette table. With the default order, the smallest values (the shallowest depths) get the first color of the palette.

−D, **−−color−flip**
Reverse the color palette.

−Y, **−−no−nice−interval**
Stretch the colors linearly over the data range instead of using rounded color intervals.

−S, **−−stretch−color**
Set the color boundaries from the distribution of the plotted values (**mbhistogram**), so that each color covers the same number of values.

−Z, **−−zbounds** *min/max*
Range of the colored quantity: depth for color modes 1-3, amplitude for mode 4, sidescan for mode 5. Default: the data range.

−A, **−−shade** *magnitude*[*/azimuth* | */zero_level*]
Shading parameters in one argument, as in **mbm_plot −A**: magnitude and illumination azimuth for mode 2, or magnitude and the amplitude of zero shading for mode 3 (**mbswath −G**). The separate options below take precedence.

−−magnitude *val*
Shading magnitude. Defaults: 2.5 for mode 2; for mode 3, 2/(amax − amin) from the amplitude range, or 1 with **−−stretch−shade**.

−−azimuth *deg*
Illumination azimuth for mode 2 [0].

−−zero−level *val*
Amplitude giving zero shading for mode 3. Default: amin + 0.2 (amax − amin), or 191 with **−−stretch−shade**.

−−shade−flip
Reverse the shading: for mode 2 the illumination azimuth is turned by 180 degrees; for a stretched mode 3 shade the gray scale is reversed.

−−stretch−shade
Shade mode 3 maps with a gray scale fitted to a Gaussian distribution of the amplitudes (**mbhistogram −G**), written to *root*.shade.cpt and passed to **mbswath −N**.

−C, **−−contour** [*cont_int/color_int/tick_int/label_int/tick_len/label_hgt*]
Plot bathymetry contours (**mbcontour −A**) at *cont_int*, with heavier contours every *color_int*, downhill ticks every *tick_int* and labels every *label_int*; *tick_len* and *label_hgt* are in inches. Missing fields get the defaults of **mbm_plot**. With no argument, the contour interval is chosen from the depth range, and the heavier contours, ticks and labels are every fourth contour (or omitted when a swath image is also drawn).

−−contour−algorithm *n*
Contouring algorithm (**mbcontour −Z**). Default: chosen by **mbcontour**, except 1 for formats 41, 101 and 102.

−−contour−nplot *pings*
Number of pings contoured at a time (**mbcontour −N**).

−N, **−−navigation** [*time_tick/time_annot/date_annot/tick_len* | **F** | **FP**]
Plot the ship track (**mbcontour −D**), with ticks every *time_tick* hours, time annotations every *time_annot* hours and date annotations every *date_annot* hours; *tick_len* is in inches. With no argument only the track is drawn. **F** annotates the track with file names as well (**FP** perpendicular to the track), with ticks every 0.25 hours.

−−nav−names [*height*[*/P*]]
Annotate the track with file names of this height in inches [0.15]; */P* places them perpendicular to the track.

−−ping−ticks [*tick/annot/len*]
Mark ping numbers along the track (**mbcontour −M**). Default 50/100/0.1.

−−footprint *factor/mode/depth*
Beam footprint parameters (**mbswath −A**). Default 1.

−−swath−scale *mode/scale*[*/min/max*]
Amplitude or sidescan scaling (**mbswath −D**).

−−begin−time *yr/mo/da/hr/mn/sc*, **−−end−time** *yr/mo/da/hr/mn/sc*
Plot only data inside this time window.

−−speedmin *speed*
Minimum ship speed in km/hr.

−−timegap *minutes*
Time gap threshold.

−−lonflip *−1|0|1*
Longitude range: −1 for −360 to 0, 0 for −180 to 180, 1 for 0 to 360. Default 0.

−T, **−−coast**
Plot a coastline (**gmt coast**). If no coastline fill, pen, river or boundary option is given, land is filled with gray 200 and the coastline drawn with a 1p pen.

−−coast−resolution *res*
Coastline resolution: f, h, i, l or c. Default **f**.

−−coast−dry−fill *color*, **−−coast−wet−fill** *color*, **−−coast−lake−fill** *color*
Fill land, water or lakes.

−−coast−river *spec*, **−−coast−boundary** *spec*, **−−coast−pen** *pen*
Draw rivers (**gmt coast −I**), national boundaries (**−N**, repeatable) or the coastline (**−W**).

−−**xy−overlay** *file*[*:fill=color*][*:symbol=symbol*][*:pen=pen*]
Plot the lon/lat data in *file* with **gmt plot**, as lines (with a pen) or symbols; repeatable. Example: "stations.xy:symbol=c0.1:fill=red".

−−**text** *lon/lat/size/angle/font/justify/text*
Place a text label; repeatable. Example: "−11.7/36.8/12/0/1/LB/Gorringe Bank".

−−**gmt−default** *param/value*
Override a GMT default parameter in the generated script; repeatable.

EXAMPLES

A shaded relief map of the swath bathymetry in a datalist, with contours and an annotated ship track, rendered at once to a PNG:

```
mbpy_plot −I datalist.mb−1 −G2 −C −N 0.25/1/4 \
  −−format png −X −V
```

A sidescan map of one file with an equalized grayscale stretch:

```
mbpy_plot −I 0003_20111219_180559_METEOR_EM122p.mb59 −G5 −S
```

A track map with file names, a coastline and a distance scale at the map center:

```
mbpy_plot −I datalist.mb−1 −N F −T \
  −−length−scale "gcenter+c36.8+w5k"
```

SEE ALSO

mbm_plot(1), **mbswath**(1), **mbcontour**(1), **mbpy_grdplot**(1), **mbpy_grdtiff**(1), **mbinfo**(1), **mbhistogram**(1), **mbsystem**(1)

BUGS

Please report any bugs to the MB-System project on GitHub (https://github.com/dwcaress/MB-System).